

Mahir Rana

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EDUCATION

Rutgers University – New Brunswick | Honors College

B.S. in Computer Science; Minors in Korean and Critical Intelligence Studies

Expected May 2027

GPA: 3.54 / 4.0

Cornell University

Machine Learning Foundations Certificate

Jun 2026 – Jul 2026

EXPERIENCE

Software Engineer

HazingInfo | University of Washington

Dec 2025 – Present

Seattle, WA

- Serve as the sole engineer on a team of 5 building a Django and PostgreSQL platform that finds hazing transparency reports from about 1,480 colleges and loads extracted incidents into the first structured incident database.
- Redesigned the workflow after feedback showed extraction alone was inaccurate, requiring human approval of every incident.
- Simplified reviewer screens for nontechnical users; deployed on Oracle Cloud with Docker Compose, Caddy, and GitHub Actions CI/CD.

Global Procurement Intern

Johnson & Johnson

Sep 2026 – Present

New Brunswick, NJ

- Research global legislation and geopolitical trends in key markets to support sourcing decisions.
- Collaborate with the Procurement, Global Services, and Policy teams to develop a reusable benchmarking framework.
- Analyze a digital accessibility policy change and supplier-facing guidance to extract supplier requirements, map who approves, applies, and must be considered, and summarize supplier impact for my team lead.

IT Consultant

Office of Information Technology | Rutgers University

May 2025 – Present

New Brunswick, NJ

- Resolve 50+ ServiceNow tickets for students, faculty, and staff across 3+ campus computer labs and libraries.
- Diagnose hardware, OS, and network issues across macOS, Linux, and Windows for users of varying technical backgrounds.

Undergraduate Research Intern

BISLab | Hanyang University

Jun 2025 – Jul 2025

Seoul, South Korea

- Built a React data table over 17,000+ post-translational modification (PTM) entries with search and per-column filters.
- Implemented a rows-per-page selector with server-side pagination, loading only each page's data to keep the table responsive.
- Cached recent query results in the backend, achieving an average query time under 3 seconds.

PROJECTS

Comms | Swift, SwiftUI, TypeScript, Firebase (Firestore, Cloud Functions, FCM), APNs

- Shipped 14 TestFlight builds of a real-time event coordination iOS app in SwiftUI and Firebase across 9 feature modules.
- Architected about 40 TypeScript Cloud Functions and 416 lines of Firestore rules for group isolation and role permissions, with sensitive actions enforced server-side.
- Built role-targeted push fan-out over Firebase Cloud Messaging (FCM) and Apple Push Notification service (APNs) with urgent alerts and server-synced Live Activities.
- Prevented races and duplicate alerts using Firestore transactions (first-tap-wins shifts, swap re-checks), per-user cooldown documents for rate limiting, and idempotent claims in a per-minute scheduler.
- Implemented fuzzy guest-name matching for door check-in using Unicode normalization, token sorting, and dynamic-programming edit distance with length-scaled thresholds.
- Wrote about 170 emulator-backed tests and root-caused a silent APNs failure, adding per-send delivery logging.

CHTR Archive & Review Dashboard | Python, Django, HTMX, PostgreSQL, Gemini API, Cloudflare R2, Docker

- Built a staged pipeline (collect, prepare text, extract, verify, publish) parsing PDFs and web pages with the Gemini API, a table parser, and OCR for scanned files.
- Archived sources in Cloudflare R2 and stored every extracted fact and human decision in PostgreSQL, so the catalog rebuilds with identical output when inputs are unchanged.
- Scored candidate URLs with a heuristic that identified the correct report 86% of the time across 4,132 documents.
- Built a Django and HTMX review queue with item locking to prevent concurrent edits and an audit trail of every decision.

Contract Risk Review Pipeline | Python, Pandas, Jupyter Notebook

Break Through Tech AI Studio × Accenture

- Built Pandas dataframes from CUAD (510 contracts, 13,823 spans, 41 categories) for an Accenture challenge.
- Validated span boundaries against source text and detected duplicate contract groups to prevent train/validation leakage.
- Shortlisted 10 clause categories into 5 review bundles (20+ positive contracts each); ranked keywords with log-odds ratios.

TECHNICAL SKILLS

Languages: Python, Java, Swift, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Django, React, Node.js, Next.js, SwiftUI, HTMX, Tailwind CSS, Pandas

Cloud & Databases: PostgreSQL, Firebase (Firestore, Functions, Auth, FCM), Google Cloud, Cloudflare R2, Docker, Oracle Cloud, Linux

Developer Tools: Git, GitHub Actions CI/CD, REST APIs, Firebase Emulator Suite, Unit/Integration Testing, Jupyter

AI/ML: PyTorch, Scikit-learn, Hugging Face Transformers, NumPy, Gemini API, AI-Assisted Development (Claude Code)